

Solution Requirements

Date	3 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users log in with a username and password; credentials are stored using salted password hashing (bcrypt) and sessions are managed via Flask-Login. Bank analysts and compliance officers use employee SSO credentials, while customers use email/OTP-based verification for self-service access.	High
2	Authorization levels	Four roles are defined: Admin (manages models, data, and user accounts), Bank Credit Analyst (submits applications, views individual predictions), Compliance Officer (runs batch screening, views high-risk reports), and Customer (self-service eligibility check limited to their own data).	High
3	External interfaces	The system integrates with the IBM Watson Machine Learning API for cloud-hosted model inference, connects to the applicant data store (historical applications CSV/database) for training and lookup, and can optionally interface with the bank's core banking system for real-time applicant records.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
4	Transactions processing	Each application submission triggers the feature engineering pipeline, is scored by the best-performing trained model (chosen among Logistic Regression, Random Forest, XGBoost, and Decision Tree), and returns an approval/rejection decision instantly. Every transaction is logged with a timestamp, input features, and result.	High
5	Reporting	The system generates dashboards showing approval/rejection rates, lists of flagged high-risk applicants, and model performance metrics (accuracy, precision, recall, F1-score), with the ability to export results as CSV or PDF for compliance review.	Medium
6	Business rules, etc.	Applicants with past-due payment records are automatically flagged as ineligible. Minimum income, employment duration, and credit inquiry thresholds are checked before the ML scoring step. Multi-class payment status codes are converted into binary eligibility labels during feature engineering.	High
7	Compliance to laws or regulations	The system adheres to applicable financial data protection regulations (e.g., RBI data-handling guidelines and GDPR-aligned data privacy practices), stores personally identifiable information securely, and maintains an audit trail of every approval/rejection decision for regulatory review.	High
8	Other	The system supports periodic retraining of the ML model as new applicant data becomes available and maintains audit logging of every prediction request and response for traceability.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The prediction pipeline (feature engineering + model inference) must return an approval/rejection result quickly enough for real-time use in the web application.	Prediction response returned in under 2 seconds per request
2	Scalability	The architecture (Flask app + IBM Watson ML deployment) must handle both single-applicant requests and large batch screening jobs submitted by compliance officers without degrading performance.	Supports batch screening of 10,000+ applicant records and 500 concurrent user sessions

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	All sensitive applicant financial and personal data must be protected at rest and in transit, with strict role-based access control enforced across all endpoints.	AES-256 encryption at rest, TLS 1.2+ in transit, role-based access control on all APIs
4	Reliability & Availability	The system must remain available for credit analysts and customers, with a documented fallback if the IBM Watson ML endpoint is temporarily unreachable.	99.5% uptime per month; automatic fallback to a locally cached best model on API failure
5	Usability & Accessibility	The web application must be intuitive for both technical users (analysts, compliance officers) and non-technical users (customers performing self-service eligibility checks).	Complies with WCAG 2.1 AA standards; core prediction flow completed in 3 clicks or fewer
6	Other	Maintainability: the Flask codebase keeps feature engineering, model training, and deployment logic in separate modules for easy updates. Portability: the application is containerized for deployment across IBM Cloud or other cloud providers.	Modular codebase with independently testable components; Docker image builds and deploys without code changes across environments